CIE-3 | AI AGENT DESIGN CHALLENGE

NAAC Information Bot

Project report and working-agent demonstration brief

PROJECT PURPOSE
A guided information assistant that helps higher-education teams organize questions about NAAC accreditation preparation, criteria, evidence, and assessment readiness.

1. Problem statement and objectives

Institutions preparing for accreditation must interpret requirements, gather evidence, coordinate owners, and keep submissions consistent. The information is spread across framework-specific documents and portal instructions; advice that ignores the institution category or current cycle can mislead users. The bot is designed to make the first step easier by answering common preparation questions, structuring follow-up, and directing users back to current official guidance.

Objectives

• Provide a simple question-and-answer interface for common NAAC preparation topics.

• Help users turn broad tasks into actionable evidence and ownership checklists.

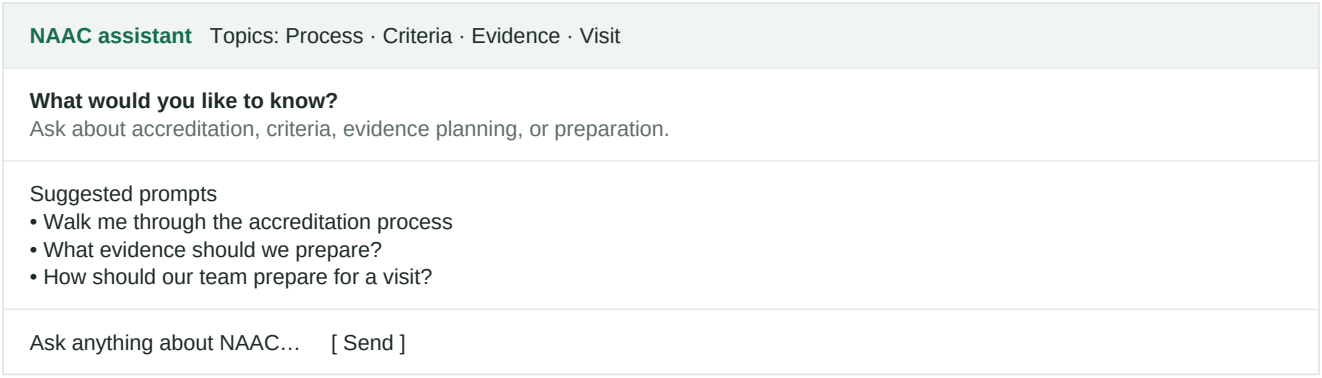
• Ask for institution type and accreditation stage when a question needs context.

• Show uncertainty and official-source reminders instead of presenting generic guidance as binding requirements.

Intended users

Internal quality assurance teams, administrators, faculty coordinators, and project teams that need a starting point for navigating accreditation preparation. The prototype is not an official NAAC service and does not determine eligibility, scores, or accreditation outcomes.

Interface preview



Schematic interface preview, not a captured runtime screenshot. The companion HTML demo is interactive and runs locally in a browser.

2. AI agent design

The proposed production design is a retrieval-augmented assistant. It should retrieve from a controlled collection of current NAAC publications, answer only from relevant passages, cite the source and version, and ask a clarifying question when institution type or cycle changes the answer. The included demonstrator uses a small local rule-based response set; it does not call an LLM, retrieve live documents, or claim current official status.

Agent workflow

1 Ask User question + optional institution context	**2 Route** Topic intent and missing context	**3 Retrieve** Approved NAAC source set (production design)	**4 Respond** Grounded answer, citation, caveat

Guardrails

• No invented deadlines, eligibility determinations, grades, or guaranteed outcomes.

• Ask users to specify institution type and cycle before giving category-specific guidance.

• Use citations with document title, issuing body, publication date/version, and a working official link.

• Escalate ambiguous, missing, or conflicting source passages instead of guessing.

• Keep institutional data private; the current demo sends nothing to a server.

Knowledge and source strategy

Production knowledge should be versioned by institution category and accreditation cycle. Each chunk should retain its official URL, title, effective date, and section identifier. Retrieval should prefer current applicable material; responses should show citations close to each claim. A maintainer should review new source versions before they enter the index.

3. Working AI agent

The companion file **naac-information-bot.html** is a standalone, offline interface demonstration. It supports suggested prompts, free-text questions, local topic matching, a new-conversation action, and caveats in every answer. Its five sample interactions are shown in section 4. It is a working interaction prototype, not a deployed generative AI agent: no LLM, NAAC document retrieval, evaluation pipeline, authentication, or hosted agent URL is configured.

Run the demo

Open the companion HTML file in a modern browser. No installation or account is required. The current response set covers broad process, criteria, evidence, and visit-preparation questions. To meet a full deployment interpretation of the assignment, connect an approved LLM and a verified, cycle-specific NAAC source collection, then capture screenshots of that running agent and provide its accessible link.

4. Sample conversation

The exchanges below demonstrate the local prototype behavior. Answers are intentionally high-level and include a reminder to verify details against current NAAC publications.

1. Explain the accreditation process.
A typical preparation path is to confirm the applicable framework, organize institutional data and evidence, complete required portal submissions, and follow the assessment steps and deadlines specified for that cycle.
Verification: Check current eligibility, process, and deadlines in official NAAC guidance.

2. What are the assessment criteria?
First identify the framework that applies to the institution and cycle. Map each applicable criterion or metric to an owner and supporting records.
Verification: Exact criteria and evidence rules depend on the current manual for the institution category.

3. What evidence should we prepare?
Build an evidence register with the requirement, source record, responsible owner, date range, and verification status. Keep submitted values traceable to source records.
Verification: This is a preparation method, not an exhaustive official document list.

4. How should we prepare for a visit?
Prepare a shared evidence repository, assign an owner for each topic, and ask teams to verify that submitted information matches underlying records and implemented practices.
Verification: Visit procedures and schedules can vary; follow current cycle instructions.

5. Can you confirm our eligibility and likely grade?
The demo does not determine eligibility or predict an accreditation grade. Specify the institution type and consult the current official framework and portal instructions for eligibility requirements.
Verification: Decisions and outcomes must be based on official NAAC processes.

5. Submission checklist and next steps

Deliverable	Status in this package
Problem statement and objectives	Included in section 1.
Agent design and safeguards	Proposed RAG architecture and guardrails in section 2.
Working agent and screenshots/link	Offline HTML demo included. No hosted link or runtime screenshot; schematic preview is labeled as such.
Sample conversation (minimum 5 interactions)	Five user questions with responses in section 4.
Single PDF report	This document consolidates the report sections.

Presentation and implementation next steps

Before submission, replace the local response set with a grounded agent connected to verified, current NAAC materials; test answer accuracy and citation behavior; capture screenshots from the running agent; and add its accessible link if hosted. Fill in student/team details and confirm the instructor accepts an offline demonstrator for the working-agent criterion.

CO-PO and Bloom mapping shown in the brief

The assignment poster maps CO4 to applying generative AI and large language model concepts, and CO5 to designing single-agent or multi-agent systems. Its Bloom mapping lists apply for problem statement and agent design, analyze for the working agent and sample conversation, and evaluate for final report/presentation. These mappings are reproduced as assignment context, not independently assessed here.

References and project notes

This report keeps institutional requirements intentionally general because the correct guidance depends on institution category and accreditation cycle. Confirm requirements in the latest official material before relying on an answer.

Official starting points

1. NAAC Basic Accreditation Portal, official portal for participating institutions:
https://assessmentonline-basic.naac.gov.in/

2. NAAC Basic Accreditation Portal, Engineering manual landing area:
https://assessmentonline-basic.naac.gov.in/mcdengg/

These are starting points only. Select the applicable institution category and current cycle materials from the portal. The prototype does not embed or reproduce a NAAC manual.

Technical scope

The HTML demonstrator is self-contained and client-side. It does not send questions to an external service or store conversation history after the page is closed. The response logic is a small topic matcher and should be treated as a UI and conversation-flow prototype.

Academic integrity and responsible use

The design encourages source verification, privacy-aware handling of institutional records, and clear disclosure of uncertainty. A production version should be evaluated using representative questions from each user group, a reviewed answer key, citation checks, and tests for unsupported claims before institutional use.

Project files

Companion demonstrator: **naac-information-bot.html**
Report: this PDF.

Submission readiness
The report and offline interaction prototype are prepared. The hosted-agent link and real runtime screenshots remain unavailable until the agent is connected and launched in an approved environment.